DESIGN OF A PUBLIC-HEALTH TOOL

OralCheck: a free, private oral cancer risk questionnaire with weights drawn mainly from published odds ratios

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oralcheck.org
Free, no account, nothing stored

Background

Oral cavity and pharynx cancer is far more survivable when found before it spreads, yet most cases are found later.

60,480
estimated new US cases, 2026

13,150
estimated US deaths, 2026

88.7%
5-year relative survival while still **localized**

36.0%
once spread to **distant** sites

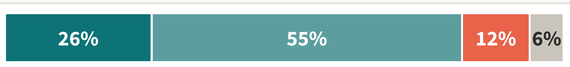


Figure 1. Cases by SEER *summary* stage at diagnosis, 2016–2022 [14].

The gap is one of timing, not treatment. A dentist can examine the mouth in minutes, but people who do not know their risk factors have little reason to ask. The term also covers two diseases: **oral cavity** cancer, driven by tobacco, alcohol and betel quid, and **oropharyngeal** cancer, driven largely by HPV-16 [3, 10]. One dental exam covers both.

Objective

Design a free, brief, private questionnaire that identifies which published risk factors a person carries, ranks them on a transparent scale derived from the literature, and turns the result into one action: **raise it with a dentist.**

It is a risk-assessment questionnaire, not a screening test. The screening test is the clinical exam; the tool's job is to make that exam more likely.

Design

- **12 questions**, about two minutes, no account; English, Spanish and Portuguese.
- Scored **in the browser**; answers are not stored.
- Output is a **tier and a next step**, never a probability, which would need a model validated on outcomes.

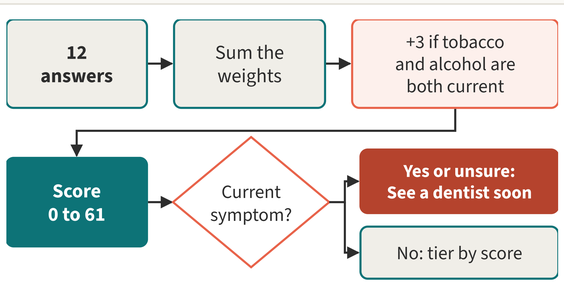


Figure 2. Scoring. A symptom adds points and also overrides the tier, since it suggests possible existing disease, not exposure [5].

How weights are set

$$\text{weight} = \text{round}(\ln(\text{OR}) \times k)$$

k = 4.47, chosen so daily tobacco, OR 6.0 (top of its 2.5–6.0 range [1]), scores 8

Summing log-odds treats risk factors as multiplying. Example: daily alcohol, OR 3.0 [2]: ln 3.0 × 4.47 = 4.91, so **5** points.

Interaction. Pooled INHANCE data found the joint effect of tobacco and alcohol greater than multiplicative, interaction parameter 2.15 [8]. round(ln 2.15 × 4.47) = **+3** when both are current. The oral-cavity value, 3.09, would give 5, so 3 is the conservative choice.

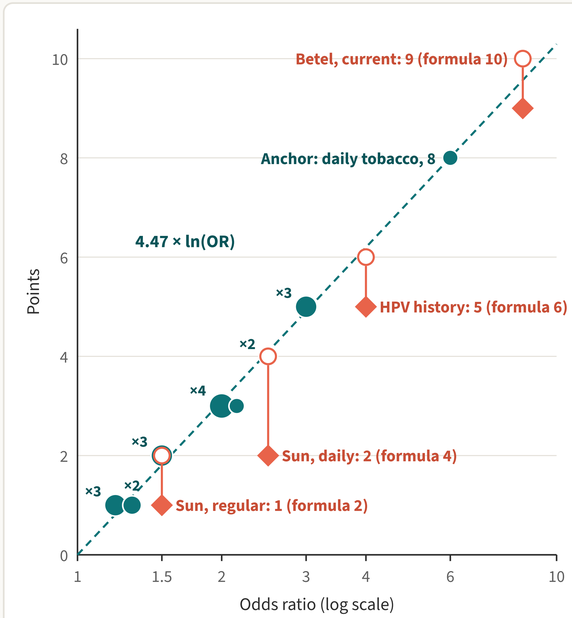


Figure 3. Assigned points against odds ratio, ranges at their midpoint. **Teal**: on the formula (×*n* = answers sharing a point). **Coral**: formula value (ring) and the weight set by judgment (diamond).

Where the formula is not followed

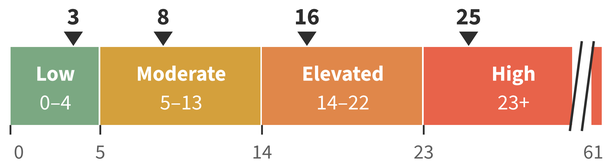
- **Betel quid, current: 9**, from the low end of its 7–10× range; the midpoint gives 10 [4].
- **HPV history: 5**, not 6, because vaccination and history are behavioural proxies, not serostatus [3, 10].
- **Lip sun exposure: 2 and 1**, half the formula's 4 and 2, set by judgment because the OR is for lip cancer alone [11].
- **Not from an OR at all:** dental-visit recency (detection proxy), "sex not stated" (population average), and the symptom answers (override).

Scoring table

Age	<35 **0** · 35–54 **2** · 55–64 **4** · 65+ **6** OR 1.5 / 2.5 / 4.0, adjusted [7]
Sex at birth	Male **3** · Female **0** · Not stated **2** OR 2.0 (SEER ratio 2.6)* [7]
Tobacco	Daily **8** · Occasional **5** · Former **2** · Never **0** OR 6.0 / 3.0 / 1.5 [1]
Alcohol	Daily **5** · Weekly **3** · Rarely **1** · Never **0** OR 3.0 / 2.0 / 1.2 [2]
HPV	History **5** · Unvaccinated **2** · Unsure **1** · Vaccinated **0** OR 3–5 / 1.5 / 1.2, proxies [3, 10]
Lip sun exposure	Daily **2** · Regular **1** · Minimal **0** OR 2–3 / 1.5, lip only [11]
Current symptom, 2+ wk	Yes **6** · Unsure **3** · No **0** Override, not an OR [5]
Family history	Close **3** · Distant **1** · No / unsure **0** OR 2.0 / 1.3 [9]
Immune or radiation	Yes **5** · Unsure **1** · No **0** OR 2–4 blended / 1.2 [12, 13]
Fruit and vegetables	Daily **0** · Weekly **1** · Rarely **3** OR 1.3 / 2.0 (low intake) [6]
Last dental visit	<1 yr **0** · 1–3 yr **1** · 3+ yr **2** · Never **3** Detection proxy, not an OR [14]
Betel quid / paan	Current **9** · Past **4** · Never **0** OR 7–10 / 2.5 [4]
Tobacco and alcohol	Both current **+3** Interaction parameter 2.15 [8]

Table 1. Maximum 58 + 3 = **61**. Coral set by judgment; dashed not OR-derived.
*Below the SEER rate ratio because tobacco and alcohol are scored separately [7, 14].

Risk tiers



3 Male sex alone: **Low**
8 Daily smoker alone: **Moderate**
16 Tobacco + alcohol + 3: **Elevated**
25 Betel + tobacco + alcohol + 3: **High**

Override: a current symptom, yes or unsure, returns the top result, "See a dentist soon", at any score.

Figure 4. Tiers and reference profiles, 0 to 61 scale (axis broken above 30).

Cutoffs were **set by judgment** against these profiles, not derived. They stayed put when the maximum rose from 53 to 61, so no unchanged profile looks safer.

Clinical review

Reviewed by a board-certified oral and maxillofacial pathologist (August 2026) and a head-and-neck surgeon (September 2026). What changed:

- Added the **immune and radiation question** [12, 13].
- Added **gum growth and tooth mobility** to symptom guidance.
- Added a caveat that many cancers occur with **no conventional risk factor**: a low score with a symptom still means go.
- Stated the **pooled-populations** limitation and the **provenance of the tier cutoffs**.
- **Oral cavity vs oropharynx**: results now name which site a person's factors point to, and say when an HPV-driven profile is likely understated. A split score was rejected: it would rest on one question.

Limitations

- **Not validated** against outcomes: no sensitivity, specificity or ROC. The score ranks; it is not a probability.
- ORs come from **different populations** and are applied to anyone; transportability is untested and is the largest uncertainty.
- One score covers **two diseases**; the HPV weight is a blended compromise.
- **Self-report**: tobacco and alcohol are often under-reported.
- The **anchor** OR, 6.0, sits above pooled estimates for current smoking and oral cavity cancer.

Next steps

- **Validation** needs outcome data and a **biostatistician**; clinical review cannot establish it.
- A **prospective pilot in dental offices**, pairing the questionnaire with the clinical exam.
- **Clinician re-review** of the weights that depart from the formula, and of the tobacco anchor.

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